

Certified but Compromised: Breaking and Fixing Lightweight Proofs of Inference

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Abstract

zkSNARKs can prove that a cloud-hosted model’s answer came from the agreed weights, but take minutes per query on large models. Anchuri et al. instead let the client re-check one random path through a committed computation, which catches a provider running a different model. We show that one tampered neuron, with the rest computed honestly, passes this check 99.8% of the time on an MNIST network and, on the real OPT-6.7B, changes the next token of every prompt we tried. No sampling rule that ignores the claimed values beats uniform sampling. Our defence checks every layer with Freivalds’ algorithm against a Reed–Solomon/Merkle commitment to the weights. It needs no SNARK, TEE or trusted setup and, given an honest commitment, accepts a wrong output with probability at most $2^{-\lambda}$. All 2,895 attacks on a basic version and all 4,973 on an optimised one were rejected. The optimised prover is one to four orders of magnitude faster than published zkSNARK provers and about 48× faster than zkLLM’s code on our GPU (Llama-2-7B, 2,048 tokens). Run non-interactively, the optimised protocol also verifies faster and sends less than zkCNN on LeNet-5 (our 8 threads against its one core) and than DeepProve’s transparent configuration on a 64-token GPT-2 prompt. DeepProve’s proof covers every position and ours the next token, and its trusted-setup configuration sends 9.4 MB. Our remaining cost is a proof that grows with the activations.

Keywords

verifiable inference, proof of inference, integrity attacks, Freivalds’ algorithm, zkML

1 Introduction

Consider a company that pays a cloud provider to serve a licensed or fine-tuned language model to its users. Nothing in a response shows that the provider ran that model with the agreed weights rather than a cheaper quantised copy, a model with a few edited weights, or one that misbehaves on certain inputs. The company cannot re-run the model to find out, since it lacks the weights and the hardware; that is why it outsourced inference. A *proof of inference* closes this gap. With each answer the provider sends a proof, and the client (here, the company) runs a check, much cheaper than inference, that fails with high probability unless the answer was computed with the agreed weights on this input.

zkSNARKs [6, 13, 20, 24, 26] prove that the answer is correct and can keep the weights secret. Their proofs are usually small and fast to check, but producing one takes far longer than inference; zkLLM needs about ten minutes on an A100 for one Llama-2-7B

prompt [26]. Anchuri et al. [2] proposed a much lighter approach. The provider commits to the full execution trace with a Merkle tree, and the client checks the computation along *one random path* from an output neuron back to the input, against a commitment to the weights. Proving drops from minutes to milliseconds. A different model produces a trace that differs from the correct one in many places, and a random path is likely to hit one of them. Anchuri et al. accordingly prove *other-model* soundness, against a provider that honestly runs a different model. They note that one path cannot catch a single changed node with probability above $1/N$ in a layer of width N , and they leave *full* soundness, against any wrong answer, open for LLMs.

We reimplemented the protocol (no code was released), reproduced its main claims, and asked what a provider can do if it keeps the committed model and changes only what it needs to. It can overwrite *a single neuron* and recompute everything after it honestly (Figures 1a and 1b). The trace is then inconsistent at one node, which a random path visits with probability $1/N$, so almost every tampered answer is accepted. Applied only to inputs that carry a trigger, the edit is a backdoor in the serving code, which leaves the committed weights clean and answers clean queries bit for bit as before. The attack carries over to the real OPT-6.7B. Checking many queries does not close the gap, because the provider chooses which queries to tamper with, and tampering with k of them is caught with probability at most k/N . Nor do the fixes Anchuri et al. suggest, sampling more paths or sampling where it matters: uniform sampling is already optimal among rules that ignore the claimed values, the rules we tested that use them still miss almost every attack, and reaching 2^{-40} forces the path protocol to open essentially the whole model and trace.

This led us to a different design (Figure 1c). The client tests one random linear combination of all the neurons of every layer (Freivalds’ algorithm [12]). When it does not hold the weights, it checks the combination against a Reed–Solomon-encoded, Merkle-committed copy of them in the style of Ligerio [1], so, as in the path protocol, it stores 32 bytes per weight matrix. The model runs as an int8 network whose results are bit-identical on the provider’s GPU and on the client, which makes the test exact. A tampered value anywhere in the network is then caught with probability $1 - 2^{-\lambda}$ instead of $1/N$, and the provider’s extra work grows with the model but not with the input. At 2^{-128} our proofs are also smaller than the paths’ at 2^{-40} : catching the single-neuron attack on a 64-token Llama-2-7B prompt with probability $1 - 2^{-40}$ takes paths that open about 13.7 GB, while our proof is 329 MB. An exact check of a few

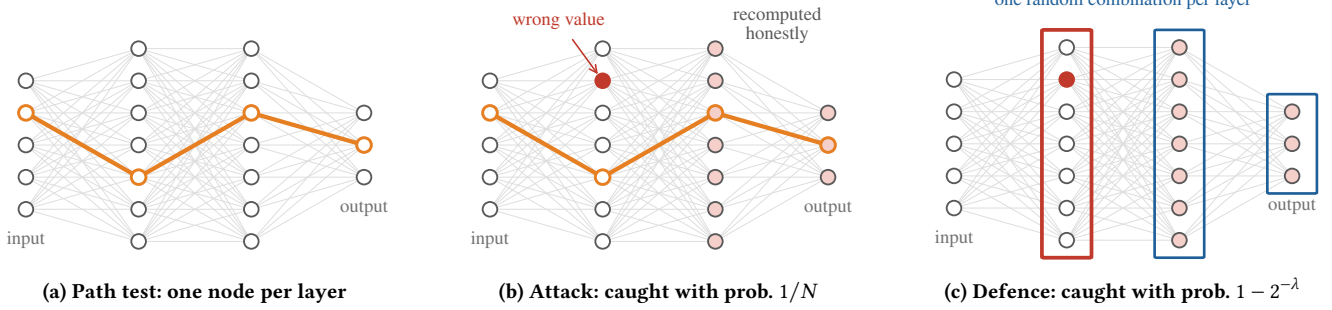


Figure 1: A single tampered neuron, with everything after it recomputed honestly, leaves one inconsistent node: a random path rarely visits it (b), while one random combination per layer catches it (c).

neurons leaves the rest of the network unconstrained; one random combination per layer constrains all of it.

Contributions.

- **An attack** that exploits the $1/N$ limit acknowledged in [2]. One tampered neuron is accepted with probability $1 - 1/N$ (99.6–99.8% on MNIST), works as a trigger backdoor that leaves the weights and clean answers unchanged, and changes the next token of 40 of 40 prompts on the real OPT-6.7B.
- **A bound on sampling-based fixes**: no sampler that ignores the claimed values beats uniform sampling, which needs essentially the whole model and trace for 2^{-40} , and a sampler that weights each parent by its contribution is blind to zeroed neurons.
- **A full-soundness protocol** for whole CNNs and transformers, with error $2^{-\lambda}$ and no SNARK, TEE or trusted setup; to our knowledge it is the first in the setting of [2]. Optimisations keep the bound and shrink the committed-weights proof 2.0–4.2× for images and prompts of up to 512 tokens.
- **A matched-model evaluation** on six image classifiers and 10 LLM architectures up to 13B parameters, against [2], 17 published systems and zkLLM’s code on our GPU. Our prover is 30–32,000× faster than the zkSNARK provers. Run non-interactively, the optimised protocol also beats zkCNN on LeNet-5 (our 8 threads against its one core) and DeepProve’s transparent configuration on 64-token GPT-2 prompts (DeepProve proves every position, ours the next token; its trusted-setup configuration sends 9.4 MB) on all three costs. Code and raw measurements are released.

2 Related Work

zkSNARK proofs of inference. A long line of work compiles neural networks into zero-knowledge proofs. For CNNs these include sum-check/GKR provers [15] such as zkCNN [20] and zkPyTorch [30], and pairing- or Plonk-based systems such as vCNN [18], ZEN [9], ZENO [10], ZKML [6], EZKL [34] and Bionetta [33]; Mystique [29] is an interactive, VOLE-based alternative. For LLMs they include zkLLM [26] (also interactive), zkGPT [24], DeepProve [13], ZKTorch [5] and Jolt Atlas [4]. They

give computational soundness and mostly small proofs, but their provers are far slower than inference: from half a second for LeNet-5 to minutes or hours for larger models.

Freivalds-style checks. SafetyNets [14] uses interactive proofs for networks with quadratic activations. Slalom [28] checks every linear layer of a GPU computation from inside an SGX enclave with Freivalds’ algorithm, in exact arithmetic modulo a 24-bit prime and optionally with a precomputed secret challenge; TwinShield [31] and VeriAttn [7] extend this to attention inside TEEs. Two recent works need no enclave. Maverick [21] combines sparse Freivalds checks with a linear code, for a client that knows the weights and draws a fresh challenge per query. LAMP [17] lets a verifier that holds only a commitment check one matrix product, using Freivalds’ check and Reed–Solomon proximity testing inside a Groth16 SNARK.

Sampling, fingerprints and optimistic protocols. Anchuri et al. [2] spot-check random paths and argue security through trace separation; SLP [19] samples which layers receive a SNARK. TOPLOC [23] and SVIP [27] fingerprint hidden states, with empirical guarantees. Verde [3], opML [8] and TAO [32] resolve disputes and assume an honest party or an active challenger.

Where we fit. To our knowledge, ours is the first protocol that gives complete CNNs and transformers full soundness $2^{-\lambda}$ in the setting of [2], where the client holds a 32-byte commitment per weight matrix, without a SNARK, a TEE or a structured reference string. Its parts are known: Freivalds’ check, exact integer arithmetic and client-side non-linear layers (Slalom, Maverick), and a Reed–Solomon-encoded Merkle commitment opened at random columns (Ligero, LAMP). We contribute the composition of these parts into a layer-chained protocol with an end-to-end soundness proof, optimisations that keep its bound, the attack, the bound on sampling and a matched-model comparison.

3 Breaking Path Sampling

3.1 Setting and threat model

A model M has a public architecture and secret or public weights. The provider acts as the prover and the client as the verifier. Before any query the client obtains a commitment C_M to the weights, for example from the model owner. As in [2], we assume that C_M is computed honestly; unlike a SNARK’s trusted setup it involves no

secret randomness, and anyone who holds the weights can recompute it. For each query x the provider returns a claimed output y and a proof, and may deviate from the protocol arbitrarily. We require *completeness* (honest answers are always accepted) and *full soundness*: any $y \neq M(x)$ is rejected except with probability ε . Anchuri et al. prove only *other-model soundness*, in which the adversary must honestly run some different model. We consider three trust settings. In setting C the client holds only C_M (32 bytes per weight matrix), as in [2] and the zkSNARKs; in K it knows the weights; and K_{pre} is K with a secret challenge precomputed once per model, as in Slalom’s preprocessing mode.

3.2 The path test and the single-neuron attack

In the protocol of [2] (RandPathTest) the prover commits to every activation a_j of the forward pass with a Merkle tree. The verifier chooses a random output neuron and walks back to the input, moving to a random parent at each layer. At every node j on the path it opens the parents’ activations and the weights w_{ij} from C_M and checks $a_j = \phi(\sum_{i \in G_j} w_{ij} a_i)$, where G_j is the set of parents of j and ϕ the activation. A path opens one weight row per layer, so the proof and the prover’s work stay small.

The single-neuron attack. Our adversary runs the correct model on x , picks one neuron j , and replaces a_j by a value that makes the output wrong (untargeted) or equal to a class of its choice (targeted). It then recomputes all later layers *with the true weights*. Every later node is consistent with its parents and every earlier node is untouched, so j is the only node whose local check fails, the only *inconsistent node*. The verifier notices the attack only if its path visits j , which happens with probability $1/N$ in a layer of width N , or $1 - (1 - 1/N)^s$ with s paths. Applied only to inputs that contain a trigger, the attack becomes a backdoor in the serving code. The prover keeps the committed weights and answers clean queries honestly, bit for bit, so no inspection of the weights reveals it and the clean-accuracy gap is zero by construction.

3.3 Why sampling does not fix it

Anchuri et al. suggest sampling more paths, or sampling where it matters instead of uniformly. Against an adversary who knows the rule, neither helps if the rule ignores the claimed values.

THEOREM 3.1 (UNIFORM SAMPLING IS MINIMAX-OPTIMAL). *Fix a layer, let U be the set of its neurons whose change can flip the output, and consider any sampler that opens at most s neurons of the layer, chosen independently of the claimed trace. Let $\pi(v)$ be the probability that neuron v is opened. Then $\min_{v \in U} \pi(v) \leq s/|U|$, with equality only if π is uniform on U .*

An adversary who knows the sampler tampers with a node of U that minimises π , so it is detected with probability at most $s/|U|$. The bound holds because a minimum is at most the average and the $\pi(v)$ sum to at most s . Constant detection therefore needs $s = \Omega(|U|)$ openings per layer, linear in the width, which is the cost the protocol was designed to avoid. Samplers that look at the claimed values are not safe either. The *contribution-weighted sampler*, for example, steps to parent i with probability proportional to its contribution $|w_{ij} a_i|$, so it never visits a neuron whose claimed activation is zero, and an adversary who flips the output by zeroing a

Table 1: Attacks on the path protocol (one path). The forgeries of [2] corrupt hundreds of the MLP’s 778 trace nodes and are always caught; ours leave 1–14 inconsistent nodes and are almost never caught.

Attack	inconsistent nodes	accepted
<i>Forgeries of [2], MLP (MNIST)</i>		
Substitute model $\tilde{M}$	≈ 574	0.000
Inverse transform	591	0.000
Logit swap	256	0.000
Gradient reconstruction	≈ 778	0.000
<i>Our attacks, MLP (MNIST)</i>		
Single neuron, layer 1	1	0.998
Single neuron, layer 2	1	0.996
Backdoor, triggered inputs	1	0.998
Stealthy (in natural range)	≈ 5	0.989
Zeroing vs. contribution-weighted	≈ 14	1.000
<i>Our attacks, OPT-6.7B (real weights)</i>		
One FFN neuron (40/40 prompts)	1	0.99994
Backdoor, target “back” (40/40)	1	0.99994

few neurons is never detected. A floor δ , sampling by $|w_{ij}|(|a_i| + \delta)$, removes this blind spot but leaves detection low (Section 3.4). Nor does a range check help, because an attack can keep every value within its neuron’s *natural range*, the values it takes on honest inputs.

3.4 The attack in practice

Reproduction. Our implementation behaves as Anchuri et al. describe. On MNIST versions of four of their substitute-model settings (three from their experiments, one from their motivation), honest provers were accepted in all 3,000 runs of each. Substitute models were caught in 99.8–100% of runs, including an 8-bit quantised copy that agrees with M on all 150 evaluated queries. Without the prescribed tolerance of 10^{-4} , an honest prover is accepted in 21–37% of runs, because a layer-wide matrix product and a single-neuron dot product round differently.

Attacks on MNIST. Table 1 gives the number of inconsistent trace nodes and the exact acceptance probability, computed by dynamic programming over the paths. The forgeries of [2] are its substitute model, the inverse-transform and logit-swap forgeries of its Appendices E and F, and the gradient reconstruction of its Section 7.2. One tampered neuron is accepted with the predicted probability $1 - 1/N$. Triggered by a 3×3 bright patch, the backdoor returns the honest output on every clean query and succeeds on every triggered query not already in the target class. It passed 99.96% of 2,300 end-to-end runs, within sampling error of the predicted 99.8%. Keeping each tampered value below its neuron’s 95th percentile takes about five neurons and still passes 99% of the time. With 250 paths, the single-neuron attack is accepted 61% of the time.

Other samplers. Against an adversary who knows the rule, uniform sampling detects 0.20% of attacks, a static importance score (the total magnitude of a neuron’s outgoing weights) 0.11%, and gradient saliency, which a verifier cannot compute without the

model, 0.016%. The contribution-weighted sampler detects nothing: zeroing about 14 of the 512 neurons flips the prediction, and end-to-end runs accepted 2,000 of 2,000 attacks. Zero is also these neurons’ most common value (65% of inputs), so activation statistics do not flag the attack. A floor $\delta = 1$ raises detection to 2% (10× uniform); other floors give 0.9–6×.

Larger models. On every CNN and every test query, changing one penultimate neuron suffices to flip the prediction. One path catches it with probability 1/84 (LeNet-5) to 1/512 (VGG, ResNet), and a change to the least-visited neuron with probability as low as 1/28 million. Reaching 2^{-40} takes 2,316–14,182 paths, which open the whole model and trace (Figure 3). On Llama-2-7B one path detects the attack with probability 1/11,008, and 2^{-40} needs about 305,000 paths (≈ 13.7 GB with the fp16 weights of [2], at 64 tokens).

A real LLM. On the real OPT-6.7B (fp16), the attacker ranks a few FFN neurons of layers 8, 16, 24 and 31 at the last position, by gradient and by their direct effect on the logits. It then finds the smallest change to one of them that changes the next token. One neuron sufficed for all 40 prompts. A fixed-target backdoor is more constrained. Because of the final LayerNorm, raising one neuron’s activation far enough forces a single token, the neuron’s *saturation token*, so the last layer’s 16,384 neurons can force only 6,187 of the 50,272 tokens. The backdoor succeeds for a reachable target (“back”: 40/40 triggered prompts, clean prompts unchanged) and fails for “hacked” (0/40). A per-neuron range check would flag its values, which exceed the neurons’ natural range, but the path protocol has none; one path detects the backdoor with probability 1/16,384, and 2^{-40} needs 454,248 paths.

4 Checking Every Layer

Our verifier tests one random linear combination of all the neurons of each weight layer. A wrong value anywhere in the layer changes this combination except with probability $1/p$, for a prime p fixed below, so the test needs no assumption about where the adversary tampers. We build it in four steps.

4.1 Construction

Step 1: an exact integer model. A random combination is a reliable test only if prover and verifier compute exactly the same numbers, so the model runs in integers. Following integer-only inference [16], each weight layer computes $Z = WX + b$, where W is int8 with one scale per output channel, X is int8 with one scale per tensor, and BatchNorm is folded into W and b . The next layer’s input is obtained by fixed-point requantisation followed by the activation or pooling, and convolutions become matrix products through im2col. On the prover’s GPU the products are computed in float32 chunks of at most 1,024 terms, so every partial sum stays below 2^{24} in magnitude and is exact, and the prover’s results are bit-identical to the verifier’s. Exactness also removes the floating-point tolerance of the path protocol, which leaves an attacker room to hide small changes.

Step 2: claims and recomputation. For every weight layer ℓ the prover sends the claimed pre-activations Z_ℓ , the *claims*. The verifier recomputes every operation that does not involve the weights from the claims of earlier layers: requantisation, ReLU,

pooling, residual additions, LayerNorm/RMSNorm, GELU/SiLU (by table look-up), RoPE, softmax, and the attention products $QK^\top$ and PV . Each weight layer’s input X_ℓ is therefore the verifier’s own computation, so the checks are chained and a prover cannot pass one layer’s check with one value and feed another into the next. What remains is to check $Z_\ell = W_\ell X_\ell + b_\ell$ for every ℓ .

Step 3: Freivalds’ check. Write $A_\ell = [W_\ell \mid b_\ell]$, a matrix with N rows (one per output neuron) and row length k , and let T be the number of columns of X_ℓ (pixels or tokens); we drop the subscript ℓ from N , k and T when the layer is clear. The verifier draws a random $\chi \in \mathbb{F}_p^N$ and checks

$$\chi^\top Z_\ell = \underbrace{(\chi^\top A_\ell)}_{u_\ell} [X_\ell; \mathbf{1}], \quad (1)$$

where we call u_ℓ the *folded row*. Let $D = Z_\ell - A_\ell [X_\ell; \mathbf{1}]$; the two sides of Eq. (1) differ by $\chi^\top D$. If $D_{im} \neq 0$, entry m of $\chi^\top D$ equals $\chi_i D_{im}$ plus terms that do not involve χ_i , so exactly one value of χ_i makes it vanish, and a wrong layer passes with probability at most $1/p$, or p^{-r} with r independent vectors. This holds however many values are wrong and wherever they are: a single tampered neuron is caught as reliably as a wholly wrong layer. Given u_ℓ , the two sides cost $O((N+k)T)$, against $O(NkT)$ for recomputing the layer; u_ℓ itself costs $O(rNk)$, paid by the prover in C and by the verifier, per query in K and once per model in K_{pre} .

We work in the BabyBear field $\mathbb{F}_p$, $p = 15 \cdot 2^{27} + 1 \approx 2^{31}$. Integers map into $\mathbb{F}_p$ without wrap-around as long as they are small, and two checks ensure this. The verifier range-checks every claim ($|z| < 2^{29}$), and setup checks that every honest pre-activation is in the same range ($128 \sum_c |w_{ic}| + |b_i| < 2^{29}$ for every row i , since int8 inputs are at most 128 in magnitude). A wrong claim and the correct value then differ by less than $2^{30} < p$, so they remain distinct in $\mathbb{F}_p$.

Step 4: committed weights. In setting C the verifier cannot compute u_ℓ , so the prover sends it. Opening t random entries of u_ℓ against the weights would be weak: a prover who changes one of the k entries escapes with probability $1 - t/k$. We therefore commit to an *encoded* copy of the weights, in the style of Ligerio [1]. At setup every row of A_ℓ is padded with zeros to a power of two $\bar{k} \geq k$ and Reed–Solomon encoded [25] to length $n_\ell = 4\bar{k}$ (Section 4.3 relaxes this), giving $\hat{A}_\ell$. The commitment C_M holds one Merkle root [22] per weight matrix, over the columns of $\hat{A}_\ell$. Encoding is linear, so $\chi^\top \hat{A}_\ell = \text{Enc}(\chi^\top A_\ell)$. The verifier encodes the claimed u_ℓ itself, opens t random columns c , and checks $\chi^\top \hat{A}_{\ell,c} = \text{Enc}(u_\ell)_c$. If u_ℓ is wrong, $\text{Enc}(u_\ell)$ and $\text{Enc}(\chi^\top A_\ell)$ are distinct codewords that agree on at most $k - 1$ of the n_ℓ positions, so t distinct columns all miss the difference with probability at most $((k - 1)/n_\ell)^t < 4^{-t}$.

4.2 Protocol and soundness

Setup happens once per model: the owner encodes and commits the weights and publishes C_M . Figure 2 shows one query, with r vectors χ and t columns per layer; besides the checks of Steps 3 and 4, the verifier range-checks the claims and checks that the recomputed layers lead to y . Because the claims precede χ and u_ℓ

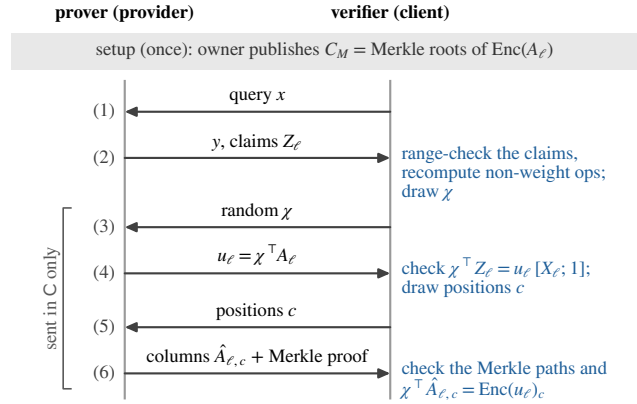


Figure 2: One query in setting C (basic protocol). The claims are fixed before χ is drawn and u_ℓ before the columns are chosen; in K and K_{pre} the verifier computes u_ℓ itself, so only (1) and (2) are sent.

precedes the column positions, the prover fixes every value before it sees the randomness that tests it.

THEOREM 4.1 (SOUNDNESS). *If C_M is honest and the prover cannot find a SHA-256 collision, then for any claimed $y \neq M(x)$ the verifier accepts with probability at most $\varepsilon = \sum_\ell [p^{-r} + (\frac{k_\ell - 1}{n_\ell})^t]$.*

Sketch. If $y \neq M(x)$, some claim is wrong; let ℓ be the first layer with a wrong claim. Its input X_ℓ is correct, because the verifier computed it from correct claims. If u_ℓ is correct, Eq. (1) fails except with probability p^{-r} (Step 3). If u_ℓ is wrong, a column check fails except with probability $((k_\ell - 1)/n_\ell)^t$ (Step 4). A union bound over the layers finishes the proof. $\square$

Parameters. For a target λ and L weight layers we set $\beta = \lambda + \log_2(2L)$, $r = \lceil \beta / \log_2 p \rceil$ and $t = \lceil \beta / 2 \rceil$; since $n_\ell \geq 4k_\ell$, each of the $2L$ terms of ε is at most $2^{-\beta}$, so $\varepsilon \leq 2^{-\lambda}$ (for VGG-16 at $\lambda=128$, $r=5$ and $t=67$). Deriving the challenges from a SHA-256 hash of the transcript, expanded with SHAKE-256 (Fiat–Shamir [11]), makes the protocol non-interactive; with 64 extra bits in β , a prover who tries up to 2^{64} transcripts still succeeds with probability below $2^{-\lambda}$.

Known weights. In K the verifier computes u_ℓ itself and messages (3)–(6) are omitted. In K_{pre} it computes the secret χ and u_ℓ once per model, so the proof is the claims alone. The prover never sees the secret χ , but it learns each verdict and, once a forgery is accepted, can repeat it. Before that, a rejection only rules out the χ that would have accepted it, so a false acceptance occurs among Q queries that reuse χ with probability at most $Q \sum_\ell p^{-r}$ ($\sum_\ell p^{-r}$ for one query).

4.3 Costs and optimisations

Costs. The prover runs the forward pass, computes u_ℓ and opens t columns, which it recomputes from W_ℓ ; its extra work, $O((r + t) \sum_\ell N_\ell k_\ell)$, does not depend on the number of pixels or tokens. A proof consists of the claims, $4 \sum_\ell N_\ell T_\ell$ bytes, which grow with the input, and of the folded rows, opened columns and Merkle hashes,

which do not. zkCNN’s and zkLLM’s proofs, by contrast, stay between tens and a few hundred kilobytes.

Optimisations. The parameters above were chosen to keep the soundness argument short, and they leave the proof larger than it needs to be. We call the protocol so far the *basic* protocol, and the one with the four changes below, checked by the engineered verifier described after them, the *optimised* protocol. Section 5 reports the optimised protocol, and Table 4 compares it with the basic one. The changes alter only how the weights are committed, which positions of the last block are proved for a next-token output, and how messages are sent. Theorem 4.1 then holds in the form

$$\varepsilon \leq \sum_j \left[p^{-r} + \prod_{i < t_j} \frac{m_j - 1 - i}{n_j - i} \right] \quad (2)$$

over the committed matrices j , with message length m_j , codeword length n_j and t_j distinct opened columns. This product bounds the chance that all t_j positions fall among the at most $m_j - 1$ where two distinct codewords agree. (i) *Columns and codes.* Each matrix opens the fewest columns that keep its term below $2^{-\beta}$, and matrices with codewords of equal length share one Merkle tree and multiproof. A public *planning rule* picks from the model’s shapes the codeword lengths that minimise the bytes per query, using at most twice as many encoded entries at setup as the basic protocol, or 2^{34} if that is more. LeNet-5’s matrices then share one tree of length 2^{18} and open 15 columns instead of 66 in each of five trees. (ii) *Transposed layout.* Committing the encoded rows $\hat{A}'$ of A^T lets the verifier combine positions instead of neurons: it draws $\chi' \in \mathbb{F}_p^T$, computes $z = Z\chi'$ and $w = [X; 1]\chi'$ itself, and checks $\text{Enc}(z)_c = (w^T \hat{A}')_c$ on t opened columns of k entries. No folded row is sent, and the opening shrinks from $rk + tN$ to tk field elements, which pays off for wide layers such as GPT-2’s output layer ($N=50,257$, $k=768$); projections of one input (query, key and value; gate and up) share one such matrix. (iii) *Decoders.* Embedding tables are committed as Merkle trees over their rows, so a wrong row needs a hash collision. For a next-token output, both parties compute the last block at the last position only. (iv) *Compact encoding.* Claims travel at 16–19 bits instead of 32 and field elements at 31; the verifier decodes before checking, and Fiat–Shamir absorbs the encoded bytes. The resulting plan (lengths, layouts, groups) follows from public shapes and is part of C_M . In K and K_{pre} , which open no columns, the optimised protocol keeps (iii), with the verifier reading the embedding rows itself, and (iv). Separately, *verifier engineering*, which changes no proof byte, makes the verifier evaluate $\text{Enc}(u_\ell)$ only at the opened positions, batch its exact products and compute attention in int32, and lets it stream the claims to a GPU. The prover of the released code also computes attention in int32, which speeds up its forward pass.

5 Results

[Red: basic-protocol placeholder, to be replaced by the second optimised run.]

5.1 Setup

Models. LeNet-5 (MNIST) and VGG-11, VGG-16 and ResNet-18 (CIFAR-10) are the CNN architectures that zkCNN, ZKML and

Bionetta report on. Some published versions are smaller variants (ZKML’s ResNet-18 has 281K parameters), so Table 5 compares only models that match ours. A 224×224 dog-vs-cat ResNet-18 trained on ImageNet classes stands in for the unreleased model of [2], and a dog-vs-squirrel one for its substitute $\tilde{M}$; the MNIST MLP (784–512–256–10) is the network of Section 3.4. We trained all of them, and int8 quantisation changed accuracy by at most 0.4 points. For language models we use the shapes of GPT-2, OPT-125M to OPT-13B, Llama-2-7B/13B and Qwen3-4B with random int8 weights. The costs depend only on the shapes, except the size of the compact encoding of Section 4.3, which depends on the values; we measure it on the same random weights, so it may differ for real checkpoints. **Every decoder up to 13B is built in full, at up to 2,048 tokens.**

Real weights. To check that the integer graph is a usable language model, we load the real OPT-125M, OPT-1.3B and OPT-6.7B checkpoints into the same graph, with one extra bias on the LM head, left over from folding the final LayerNorm into it. On 4,064 tokens of WikiText-2 their perplexity is 73.8, 37.4 and 36.0, against 64.7, 34.9 and 27.1 in fp32. These values depend on two public constants, the gain G of the integer normalisation (4, 3 and 2) and the activation-calibration percentile; the $G=32$ of our cost runs clips OPT’s outlier features and gives perplexities of 326 to 23,849. We chose both on the evaluation text, so these perplexities are slightly optimistic.

Hardware and baselines. The prover is one NVIDIA L40S (48 GB) and the verifier 8 threads of an AMD EPYC 9334; times are medians over a run’s queries. Of the 30–70B shapes only 1- and 2-block builds fit. From them we extrapolate proof sizes but not times; **where a full build exists, the extrapolated size is within 0.01% of it.** Published costs are taken as reported on their own hardware, each checked against its source; for the path protocol we compute exact detection probabilities and bytes on our graphs and take its time per path from Anchuri et al. The basic protocol was measured at commit 9401431, and a second run of it on an RTX 2080 Ti reproduced all 1,978 hardware-independent values. The optimised protocol was measured with the released code (commit c8be5eb) on the same machines; that code reproduces the basic proofs byte for byte. All tables and figures are generated from the released raw measurements.

5.2 Soundness in practice

In the optimised runs the verifier accepted all **4,380** honest queries and rejected all **4,973** attacks. The **4,950** image-model attacks, at $\lambda=40$ with committed weights, were one changed value, one changed penultimate neuron, one changed logit, 1% of the weights perturbed, **the substitute $\tilde{M}$** , and two forgeries that each pass every check but one; **3,300 of them were on commitments built under the planning rule.** Every language-model run added +1 to one middle-layer pre-activation (23 attacks, **0 of them on 1- and 2-block builds of the 30–70B shapes**), and for five models from OPT-1.3B to Llama-2-13B we also ran the image-model attacks that apply to them. Freivalds’ check rejected 3,909 attacks and the column check against the code **914**, including every forged folded row and every tampered claim of a layer in the transposed layout; the Merkle check rejected all **150** forged columns, so each

Optimised at 128 bits sends less than paths at 40 bits

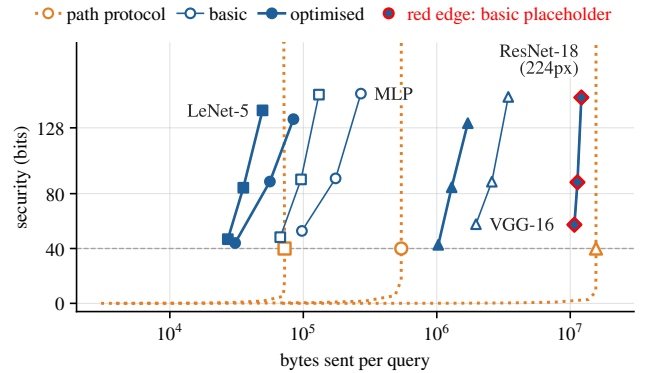


Figure 3: Security against bytes sent per query, in bits: $-\log_2$ of the probability that a wrong output is accepted (paths: the single-neuron attack; ours: the proven bound at $\lambda=40, 80, 128$; hollow: basic; shape: model). The paths reach 40 bits only by opening the whole model and trace (vertical lines).

Table 2: Image models: one query at $\lambda=128$, optimised protocol (acc.: accuracy of the int8 model). Proving takes milliseconds, and the proof is smaller than the int8 weights (one byte per parameter) for every model but the 224-pixel ResNet.

Model	Params	acc.	committed (C)			K_{pre}	
			prove	verify	proof	verify	proof
MLP (MNIST)	536K	97.7%	3.8 ms	1.7 ms	84.3 kB	0.5 ms	3.1 kB
LeNet-5 (MNIST)	62K	99.3%	5.3 ms	2.4 ms	49.5 kB	1.7 ms	26.1 kB
VGG-11 (CIFAR)	9.8M	91.1%	12.9 ms	11.9 ms	1.0 MB	11.8 ms	610 kB
VGG-16 (CIFAR)	15.2M	93.0%	17.3 ms	14.9 ms	1.7 MB	19.4 ms	1.1 MB
ResNet-18 (CIFAR)	11.2M	94.2%	20.6 ms	24.7 ms	2.3 MB	46.5 ms	2.5 MB
ResNet-18 (224px) [†]	11.2M	95.0%	27.4 ms	175 ms	12.1 MB	105 ms	9.9 MB

[†]Our ImageNet-trained counterpart of the classifier of [2].

of the three checks is needed. The basic protocol’s run accepted all 8,709 honest queries and rejected all 2,895 attacks, 804 of them on language models.

Figure 3 plots security against bytes. The optimised protocol **lands close to its target** (43–57 bits at $\lambda=40$ and 131–150 at $\lambda=128$) and, **on every model**, reaches 2^{-128} with fewer bytes than the paths need for 2^{-40} . The basic protocol overshoots its target because r and t are rounded up (48–58 and 150–153 bits), and at $\lambda=40$ it sends 1.1–8.3× fewer bytes than the paths need for 2^{-40} .

5.3 Cost

Image models. With committed weights (Table 2) the prover needs 3.8–27.4 ms per query and the verifier 1.7–175 ms. The proof, 49.5 kB–12.1 MB, is 1.3–9.3× smaller than the int8 weights for the MNIST and CIFAR models; **for the 224-pixel ResNet, whose claims grow with the number of pixels, it is about their size.**

Language models. With a 64-token prompt GPT-2 is proved in 80.2 ms and Llama-2-7B in 1.7 s (Table 3), and Llama-2-7B’s 329 MB

Table 3: Language models: next-token logits of one prompt at $\lambda=128$, optimised protocol, random int8 weights (GPU: verifier on the GPU; red -: no basic run). Proving takes under 30 s, and at 2,048 tokens the K_{pre} proof, which holds only claims, is almost the whole C proof.

Model	Params	committed weights (C)				K _{pre}	
		prove	verify	GPU	proof	verify	proof
64-token prompts							
GPT-2	124M	80.2 ms	92.0 ms	427 ms	14.8 MB	137 ms	21.8 MB
Qwen3-4B	4.0B	1.9 s	4.6 s	–	659 MB	1.6 s	284 MB
Llama-2-7B	6.7B	1.7 s	1.1 s	3.8 s	329 MB	1.5 s	349 MB
Llama-2-13B	13.0B	4.6 s	6.7 s	–	1.2 GB	2.7 s	547 MB
2,048-token prompts							
OPT-125M	125M	1.1 s	14.2 s	691 ms	733 MB	12.7 s	692 MB
OPT-1.3B	1.3B	5.7 s	78.9 s	3.0 s	3.8 GB	81.2 s	3.7 GB
OPT-6.7B	6.7B	16.2 s	2.7 min	6.4 s	10.1 GB	2.7 min	9.7 GB
Llama-2-7B	6.7B	17.6 s	2.8 min	6.9 s	11.6 GB	2.7 min	11.2 GB
Llama-2-13B	13.0B	27.1 s	3.7 min	–	18.1 GB	3.7 min	17.5 GB

Table 4: Basic $\rightarrow$ optimised protocol on the same machines (C, interactive, $\lambda=128$ unless marked). In setting C, below 2,048 tokens, the proof shrinks 2.0–4.2 $\times$ and verification speeds up 2.7–8.4 $\times$; the prover slows down only on LeNet-5.

Model (data or tokens)	prove	verify	proof
LeNet-5 (MNIST)	4.3 $\rightarrow$ 5.3 ms	7.0 $\rightarrow$ 2.4 ms	131 $\rightarrow$ 49.5 kB
VGG-16 (CIFAR)	21.5 $\rightarrow$ 17.3 ms	63.4 $\rightarrow$ 14.9 ms	3.4 $\rightarrow$ 1.7 MB
ResNet-18 (CIFAR)	26.1 $\rightarrow$ 20.6 ms	110 $\rightarrow$ 24.7 ms	4.6 $\rightarrow$ 2.3 MB
GPT-2 (64)	173 $\rightarrow$ 80.2 ms	468 $\rightarrow$ 92.0 ms	62.1 $\rightarrow$ 14.8 MB
GPT-2 (512)	163 $\rightarrow$ 134 ms	1,012 $\rightarrow$ 375 ms	213 $\rightarrow$ 89.5 MB
Llama-2-7B (1)	2.7 $\rightarrow$ 1.5 s	2,527 $\rightarrow$ 299 ms	418 $\rightarrow$ 130 MB
Llama-2-7B (64)	2.7 $\rightarrow$ 1.7 s	4.4 $\rightarrow$ 1.1 s	762 $\rightarrow$ 329 MB
Llama-2-7B (2,048) ^g	17.6 $\rightarrow$ 17.6 s	6.9 $\rightarrow$ 6.9 s	11.6 $\rightarrow$ 11.6 GB
Qwen3-4B (8) ^k	135 $\rightarrow$ 119 ms	263 $\rightarrow$ 156 ms	36.1 $\rightarrow$ 20.3 MB

Optimised: Section 4.3, with the last block at the last position for GPT-2 and Qwen3-4B. ^gGPU verifier, streaming when optimised; prover and proof from the CPU-verifier runs, as in Table 3. ^k K_{pre} , $\lambda=40$, one verifier thread, embedding look-ups.

proof is about 21 $\times$ smaller than its int8 weights. At 2,048 tokens Llama-2-7B is proved in 17.6 s, but the proof grows to 11.6 GB, since the claims grow with the prompt, and CPU verification to 2.8 min, because the verifier recomputes attention over all token pairs; in setting C a GPU verifier is 21–26 $\times$ faster there and 1.1–1.2 $\times$ at 64 tokens. Prompts whose claims are all sent before the challenge can share one set of challenges and columns; in batches of eight 64-token prompts, Llama-2-7B’s prover then needs 0.40 s and its proof 401 MB per prompt. For Llama-2-70B the extrapolated proof at 64 tokens is 3.7 GB, 18 $\times$ smaller than its int8 weights.

Figure 4 places these runs among the published ones. Part of the prover gap is our GPU, since all the zkSNARKs except zkLLM ran on CPUs. The orange cross is one path of [2], as reported on an RTX 3090.

5.4 Effect of the optimisations

Table 4 compares the two protocols on the same machines. Much of the 2.7–8.4 $\times$ verifier gain in setting C below 2,048 tokens comes from verifier engineering, which also speeds up the basic proofs

Table 5: Optimised protocol against published systems on matched models, as theirs/ours (above 1: ours faster or smaller; bold: better on all three). Our prover is faster in every row; our proof is smaller only in the bold rows and against Maverick. For the language models, ours prove the next-token logits.

System	Model (tokens)	prover	verifier	proof
zkCNN ^a	LeNet-5	81 $\times$	2.2 $\times$	1.1 $\times$
	Bionetta	690 $\times$	3.8 $\times$	1/72
zkCNN ^a	VGG-16	4,481 $\times$	3.2 $\times$	1/6.4
ZKML	VGG-16	32,334 $\times$	1/1.9	1/180
zkGPT ^b	GPT-2 (64)	239 $\times$	3.3 $\times$	1/164
DeepProve ^c	GPT-2 (64)	375 $\times$	13 $\times$	1.3 $\times$
DeepProve ^c	GPT-2 (512)	973 $\times$	3.8 $\times$	1/3.6
ZKTorch [*]	Llama-2-7B (1)	1,786 $\times$	334 $\times$	1/5.7
Maverick ^d	Qwen3-4B (8)	3.0 $\times$	1/1.2	1.8 $\times$
zkLLM	OPT-125M (2,048)	70 $\times$	1/2.0	1/5,195
zkLLM	OPT-1.3B (2,048)	39 $\times$	1/3.4	1/25,898
zkLLM	OPT-6.7B (2,048)	34 $\times$	1/3.1	1/64,338
zkLLM	Llama-2-7B (2,048)	35 $\times$	1/2.9	1/63,313
zkLLM	Llama-2-13B (2,048)	30 $\times$	-	1/96,436
zkLLM ^e	Llama-2-7B (2,048)	48 $\times$	-	-

^aVerifier on one core (ours on 8 threads). ^bzkGPT states no prompt length; ours uses 64 tokens. ^cTransparent (BaseFold) configuration, proving every position; its HyperKZG one sends 9.4 MB at 64 tokens. ^{*}Interactive (Fiat-Shamir not run). ^d K_{pre} , $\lambda=40$, one verifier thread, like Maverick’s client, whose time includes its non-linear replay. ^ezkLLM’s public code on our L40S; it reports no verifier time or proof size (-). Our verifier uses 8 CPU threads, or at 2,048 tokens the streaming GPU verifier, with prover and proof from the CPU-verifier run, as in Table 3. Red -: no basic run.

(468 to 197 ms for GPT-2 at 64 tokens). The prover gets faster too, except on the two MNIST models (4.3 to 5.3 ms for LeNet-5 and 3.1 to 3.8 ms for the MLP), where encoding the claims costs more time than the planning rule saves (3.0 and 1.9 ms without the compact encoding). At 2,048 tokens the optimised protocol proves Llama-2-7B in 17.6 s, verifies it on the GPU in 6.9 s and sends 11.6 GB, against 17.6 s, 6.9 s and 11.6 GB for the basic protocol. Without any change to the proof, the released code’s prover and the streaming verifier already take 5.4 s and 1.7 s on the basic proof format, and the prover then takes 1.9 $\times$ the plain int8 forward pass (the K_{pre} prover). The prover gains from int32 attention and the verifier from that and from streaming; we did not separate the two. On Llama-2-13B the same streaming verifier takes 9.1 s, 5.4 $\times$ Llama-2-7B’s for a 1.9 $\times$ larger model, with host memory peaking at 63 GiB, for reasons we have not found. The optimisations make the one-time setup longer (114–198 s instead of 10–11 s for GPT-2, about 8.5 instead of 4.5 min for Llama-2-7B).

5.5 Comparison with published systems

Table 5 compares the optimised protocol with published systems, each in the matching setting. Against non-interactive systems we use Fiat-Shamir (except ZKTorch, where it was not run), and against zkLLM, which is interactive itself, the interactive protocol. We use committed weights except against Maverick, whose client knows the weights and checks with its own private randomness, as K_{pre} does.

Prover. Ours is faster than every zkSNARK prover on its matched model, by 30 $\times$ (zkLLM, Llama-2-13B) to 32,000 $\times$ (ZKML, VGG-16). Part of this is hardware, so we also ran zkLLM’s public

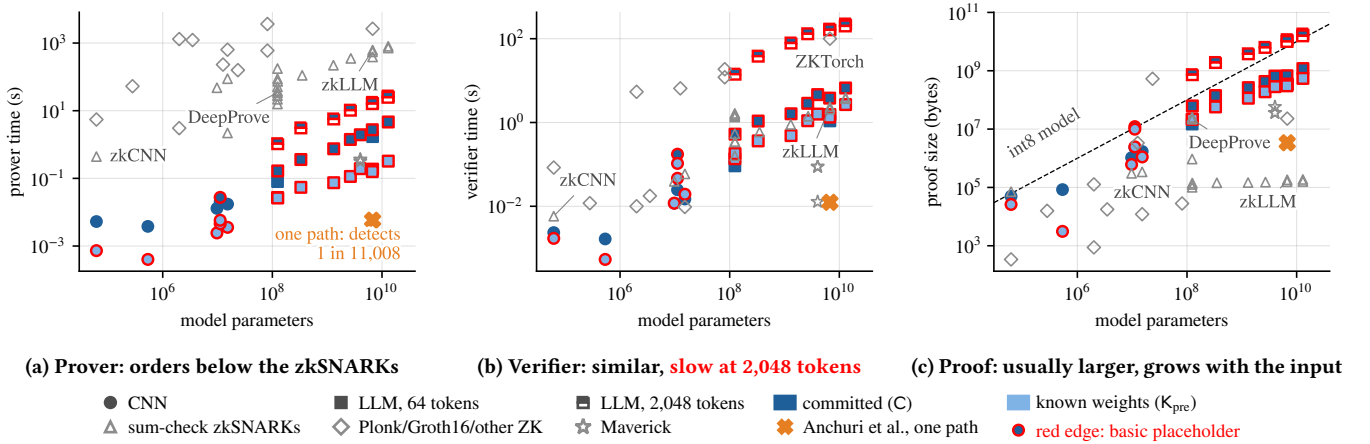


Figure 4: Cost against model size: the optimised protocol (blue; verifier on 8 CPU threads) and published systems as reported on their own hardware (grey; orange: one path of [2]). Table 5 compares them on matched models.

code on our L40S. Its demo proves the components of one layer separately and reports no verifier time, so we time one layer (median of three runs on each of two layers) and multiply by the number of layers. The demo lacks zkLLM’s cross-layer optimisation and the attention output projection, and its attention step writes no output, so we count the layer’s other skip connection twice instead of the one after it. For Llama-2-7B at 2,048 tokens this gives about 844 s, 48× our 17.6 s. We do not use the demo for Llama-2-13B, whose time per layer grows out of proportion (2.5× the 7B figure).

Verifier and proof. Our verifier is slower only than ZKML’s on VGG-16 (1.9×), zkLLM’s at 2,048 tokens (2.0–3.4×) and Maverick’s (1.2×). Maverick’s 124.5 ms is its 87.1 ms of matrix checks plus the 37.4 ms in which its client replays the non-linear operations, as our verifier does (the client’s whole online time is 260.2 ms); ours takes 155.5 ms, 43.4 ms of it decoding the compact encoding. Without the compact encoding our Qwen3-4B run beats Maverick on all three costs (117 against 354.5 ms, 111.0 against 124.5 ms and 35.19 against 36.08 MB); our prover, however, runs on the GPU and Maverick’s on 64 CPU threads. Maverick also draws a fresh challenge per query, whereas K_{pre} reuses a secret one. On LeNet-5 and on GPT-2 with a 64-token prompt the non-interactive optimised protocol wins on all three costs: 5.4 ms, 2.6 ms and 63.3 kB against zkCNN’s 441 ms, 5.8 ms and 71.3 kB, and 91 ms, 105 ms and 16.6 MB against DeepProve’s 34.2 s, 1.35 s and 21.7 MB. zkCNN’s verifier ran on one core and ours on 8 threads, and zkCNN’s protocol has a zero-knowledge variant, which ours lacks. DeepProve’s figures are for its transparent (BaseFold) configuration, whose proof certifies all 64 positions where ours certifies the next-token logits; its HyperKZG configuration, which needs a structured reference string, sends 9.4 MB. Elsewhere our proof is larger, by 3.6–180× against the other short-input proofs and by 5,200–96,000× against zkLLM’s at 2,048 tokens.

6 Discussion

Implications. The path protocol catches a provider who runs the wrong model, but not one who runs the committed model

and edits the answer, and none of the sampling rules we analysed closes that gap at sub-linear cost. Full soundness, however, does not need a zkSNARK prover: one random linear combination per weight layer gives a $2^{-\lambda}$ guarantee with no trusted setup and no assumption about how models behave. Because the verifier recomputes attention, the protocol suits a gateway or an auditor with a GPU and a fast link rather than a phone. It works best for large models with short prompts, where the optimised proof is far smaller than the model (21× for Llama-2-7B at 64 tokens).

Limitations. (i) The proof grows with the activations; at 2,048 tokens it is mostly claims and gigabytes for 7B models, against zkLLM’s 157–183 kB. (ii) The verifier’s cost grows with the square of the prompt length: at 2,048 tokens the CPU verifier needs up to 3.7 min, and the GPU verifier is 2.0–3.4× slower than zkLLM’s. (iii) The provider must serve the integer model, which changed CNN accuracy by at most 0.4 points and raised the perplexity of real OPT checkpoints by 7–33% (constants chosen on the evaluation text); the LLM costs were measured with random weights. (iv) Like the path protocol we assume an honest weight commitment, which the planning rule’s longer codewords make up to 19× slower to compute; a malformed one would need a proximity argument as in Ligerio. (v) The opened columns reveal random linear combinations of each weight row, so the protocol is not zero-knowledge. (vi) We prove one forward pass, without a KV cache. (vii) The prototype is Python, and the same-GPU zkLLM comparison rests on its demo.

Future work. The claims, which the compact encoding roughly halves, could be replaced by a sum-check argument for the non-weight operations. A native verifier and a KV cache would also help, and per-channel normalisation gains may narrow the perplexity gap at no proving cost. Before any wider publication we will share the attack with the authors of [2].

7 Conclusion

A single tampered neuron defeats the path protocol of [2], on an MNIST network and on the real OPT-6.7B, and no sampling rule

we analysed repairs it at sub-linear cost. Checking every layer with Freivalds' algorithm against a Reed–Solomon/Merkle commitment fixes it. The resulting protocol accepts a wrong output with probability at most $2^{-\lambda}$, and it rejected all 2,895 attacks on the basic protocol and all 4,973 on the optimised one. Its prover is one to four orders of magnitude faster than published zkSNARK provers. Run non-interactively, the optimised protocol also beats zkCNN on LeNet-5 (our 8 threads against its one core) and DeepProve's transparent configuration on GPT-2's next-token logits (DeepProve certifies all 64 positions; its trusted-setup configuration sends 9.4 MB) in verifier time and proof size. Long prompts, **where the proof is mostly claims**, remain its weak point.

8 Author Contributions

[To be completed by the team.] Ofek Lavi: [...]. Erel Barzilay: [...]. Edo Kadosh: [...].

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